

Fairness in Credit Scoring Models: From Algorithmic Fixes to Policy Frameworks

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Abstract

ML based credit scoring models have transformed modern lending by enabling large-scale, data-driven risk assessments. However, their increasing reliance on historical data raises deep concerns about **fairness, bias, and transparency**. This review critically examines two approaches of fairness research in credit scoring. The first group of studies focuses on modeling approaches like counterfactual fairness, empirical fairness interventions, and explanation stability while the second group turns toward data-centric and interpretability-driven technical fixes. Together, they reveal key tradeoffs between fairness, profit, and trust, highlighting that while algorithmic refinements can mitigate discrimination statistically, they cannot guarantee justice in financial decision-making.

1 Introduction

Credit scoring models are used to determine loan eligibility and pricing, affecting billions of individuals worldwide. As financial institutions adopt machine learning to automate these decisions, the risk of reproducing historical discrimination has become a pressing ethical and regulatory issue. Biased data or uncritical model design can unfairly deny credit access to disadvantaged groups, deepening economic inequalities. Governments in both the U.S. and E.U. now require “**equal opportunity by design**” in automated lending systems, underscoring the need for fairness-aware algorithms.

Despite this urgency, the gap between formal fairness definitions and practical deployment remains wide. The studies reviewed here collectively address this gap. Kusner et al. [1] introduce a causal framework for *counterfactual fairness*, emphasizing individual-level equity. Kozodoi et al. [2] empirically assess fairness interventions across real credit datasets, focusing on the trade-off between fairness and profit. Chen et al. [3]

explore how enforcing fairness can unintentionally destabilize model explanations, complicating interpretability. This review synthesizes their contributions before turning to a second wave of research that approaches fairness through better data and interpretability rather than through modeling adjustments.

2 Fairness Frameworks and Empirical Evidence

This section reviews a first strand of work that treats fairness primarily as a **modeling problem**. The papers by Kusner et al. [1], Kozodoi et al. [2], and Chen et al. [3] illustrate how different formalizations of fairness like causal, empirical, and interpretability-aware shape what counts as a fair credit scoring model in practice.

2.1 Causal Counterfactual Fairness

Kusner et al. [1] discuss algorithmic fairness by introducing a **causal framework** that defines fairness in terms of counterfactual reasoning. Their concept of **counterfactual fairness** requiring that a prediction remain unchanged if an individual’s protected attribute were altered in a hypothetical world represents a significant shift from just statistical to structural understandings of bias. This definition has strong conceptual appeal because it aligns with how discrimination operates in reality: through causal pathways rather than correlations.

However, while conceptually elegant, this framework is difficult to apply in practice. It requires knowing exactly how every factor in the data influences credit outcomes including hidden social or economic variables which is rarely possible in the real world. In simple terms, building and comparing all those “what-if” causal scenarios is not feasible with the kind of data financial institutions actually have. Nevertheless, its contribution is foundational, as it reframes fairness as a question of **causal accountability** rather than data manipula-

tion, forcing later work to confront the structural roots of bias.

2.2 Empirical Fairness Interventions

Kozodoi et al. [2] take a more practical approach by **empirically testing fairness** interventions on 7 real credit-scoring datasets. Their comparative study across pre-processing, in-processing, and post-processing techniques provides evidence on the economic feasibility of fairness. By measuring both fairness metrics and profit, they move beyond moral debates to examine business constraints which is a perspective often missing in the fairness literature.

Yet, while the results are encouraging showing that fairness can be improved without severely hurting profits, the analysis still simplifies the real trade-off between ethics and economics. Kozodoi et al. test several fairness interventions such as data reweighting, adversarial debiasing, and post-processing adjustments and evaluate their impact on both prediction accuracy and profit. However, the study focuses mainly on short-term performance metrics, overlooking broader costs of fairness interventions, such as added model complexity, regulatory challenges, or long-term effects on borrower behavior. By promoting equalized odds as the “most suitable” fairness rule, the authors also assume that fairness can mean the same thing across all contexts which is an assumption that ignores how credit access is shaped by deeper social and economic inequalities. While Kozodoi et al. provide useful evidence that fairness and performance can coexist, their framework ultimately treats fairness as a technical metric rather than a question of social justice.

2.3 Explanation Stability under Fairness Adjustments

Chen et al. [3] explore a less-discussed but important question about how fairness adjustments affect how we understand model decisions. Using explainability tools like **SHAP** and **LIME**, they find that applying fairness constraints often changes which features appear most important, making explanations less stable and consistent. In practice, this means that the reasoning behind credit decisions can shift after fairness corrections, making the model harder to audit or justify to regulators and applicants. This finding highlights a key challenge about **fairness and explainability**, usually seen as

separate goals, can sometimes conflict rather than reinforce each other.

The study’s value lies in moving the discussion from outcomes to *understandability* asking not just whether models are fair, but whether people can make sense of them. However, their experiments also expose how fragile common interpretability methods can be, without providing solutions to stabilize them. The broader takeaway is a paradox: efforts to make credit scoring models fairer may unintentionally make them harder to explain and trust.

3 Critical Synthesis and Discussion

Together, these studies show that fairness in credit scoring is not a single technical target but a balancing act between justice, profit, and trust. Kusner et al. [1] frame fairness as a **causal principle**, Kozodoi et al. [2] as a practical **tradeoff** between ethics and profitability, and Chen et al. [3] as a question of how **model interpretability**. Each adds an important piece to the puzzle but none offers a complete solution.

A key challenge that emerges is the gap between fairness in theory and fairness in practice. Individual level fairness, treating each borrower the same way is almost impossible to achieve with large, complex systems, while group-level metrics are easier to apply but often hide inequalities within those groups. Another unresolved tension is between fairness and interpretability as Chen et al. [3] show, models adjusted to be fairer can also become harder to explain, reducing transparency for regulators and users.

Most importantly, all three studies treat **fairness as a static property** of models rather than as something that evolves within financial systems. None explore how fairness holds up over time as data, markets, and behaviors change. Moving toward *sustainable fairness* will require models and policies that remain equitable even as the world they operate in shifts.

4 Beyond Model Fixes: Data-Centric and Interpretability Approaches

The studies reviewed so far largely approached fairness as a modeling problem by treating bias as something to be corrected through causal reasoning, optimized training, or explanation stability. However, fairness in credit scoring does not depend only on how models are built, but also on the data

they learn from and the transparency of their outputs. Three works : Hlongwane et al. [4], Kozodoi et al. [5], and Hurlin et al. [6] represent this idea. Each proposes a different “**technical fix**” aimed at making credit scoring fairer through better data or clearer models. Together, they illustrate both the promise and the pitfalls of treating fairness as a problem of information quality rather than institutional design.

4.1 Alternative Data for Inclusion

Hlongwane et al. [4] address the challenge faced by individuals with little or no credit history often called “thin-file” applicants who are excluded from traditional credit scoring. They argue that unfairness partly arises because models lack enough information about such borrowers. To fix this, the authors suggest using alternative data sources like phone usage, spending behavior, or regional information to build a more complete picture of financial reliability. In theory, this approach aims to make credit access more inclusive by giving the model more context.

However, this solution raises important **ethical concerns**. Many of these new data sources can indirectly reflect sensitive traits like race, gender, or income level, even if they don’t include them directly. For example, where someone lives or shops can reveal their socioeconomic background. Without proper fairness checks, adding more data could simply reproduce old biases in new forms. The study also pays little attention to privacy or consent which are issues that become critical when personal digital traces are used for financial decisions. In short, while the idea of using more data sounds fairer, it may instead make discrimination more hidden and harder to detect. This highlights a key paradox of data-centric fairness: sometimes, more data does not mean fairer data.

4.2 Bias-Aware Self-Learning and Bayesian Evaluation

Kozodoi et al. [5] address a more subtle kind of bias in credit scoring—sampling bias. Credit institutions only know whether approved applicants repay their loans; they have no data on those who were rejected. This “invisible gap” means models are trained on an incomplete picture of borrower behavior, which can distort predictions.

To tackle this, the authors propose two methods. The first, called **Bias-Aware Self-Learning** (BASL), lets the model guess what might have hap-

pened if rejected applicants had been approved and then retrains itself using the most confident of these guesses. However, this can backfire, as a biased model may simply reinforce its own assumptions. The second method, **Bayesian Evaluation** (BE), takes a safer approach by simulating multiple possible repayment outcomes using probabilistic reasoning. While this reduces bias, it still relies heavily on prior assumptions and access to detailed historical data, which are often constrained by privacy or regulation.

The authors’ results show modest improvements in fairness metrics without major losses in accuracy, but the gains remain uneven across datasets. Both BASL and BE are clever statistical tools that fill data gaps rather than challenge the structural causes of bias. By treating fairness as a statistical imbalance to be corrected, they risk optimizing models for “data completeness” rather than true equity. In short, these methods polish existing biases with more sophisticated mathematics instead of redefining what fairness should mean in credit evaluation.

4.3 Empirical Fairness Testing in Deployed Credit Models

Hurlin, Pérignon, and Saurin [6] shift the focus from designing new algorithms to evaluating how fair real-world credit scoring systems actually are. Using operational European bank data, they test demographic parity and equal opportunity, moving fairness research closer to regulatory practice.

Their analysis shows that even after accounting for legally permitted features, large disparities persist across gender, age, and regional groups. This suggests that bias often reflects deeper socioeconomic structures rather than technical model flaws.

The study reframes **fairness as accountability rather than design** but remains limited by reliance on group-level statistical metrics that can miss individual and intersectional harms. While the authors call for greater transparency and regular audits, the solutions remain compliance-oriented, promoting procedural oversight more than transformative change in how credit is allocated.

4.4 Synthesis: The Limits of Technical Optimism

Across these three papers, a clear pattern emerges. Hlongwane et al. [4] aim to make credit scoring fairer by adding more data, Kozodoi et al. [5] by

filling in missing data, and Hurlin et al. [6] by checking how existing models perform in practice. All three show that fairness measures can improve certain metrics or make models seem more balanced, but none truly prove that these changes lead to fairer outcomes for real people. Fairness is still being judged mainly through internal statistics like accuracy, profit, or parity ratios, rather than through how credit is actually distributed or experienced by borrowers.

This reflects a deeper problem in current research, **technosolutionism**, the belief that better technology alone can solve systemic issues. By focusing on fixing models instead of questioning how financial systems themselves allocate risk and opportunity, these efforts risk making bias look more refined rather than removing it. Real progress will require combining technical audits like those of Hurlin et al. [6] with stronger data governance, regulatory enforcement, and consumer protection. Without this broader accountability, credit scoring may only appear fairer on paper while continuing to reproduce the same inequalities in practice.

5 From Technical Fixes to Policy Frameworks

The studies reviewed so far reveal that algorithmic fairness in credit scoring has largely been treated as a modeling challenge, to be solved through better data, balanced learning, or more interpretable outputs. Yet, fairness cannot be sustained by technical design alone; it must also be *governed* through policy. In recent years, regulators have begun translating these ideas into practice, reframing fairness as a legal and institutional obligation rather than a purely computational goal. Two frameworks, India's **RBI Digital Lending Guidelines** and the **World Bank Fair Credit Principles** illustrate this emerging policy turn.

5.1 RBI Guidelines: Procedural Fairness in Practice

The Reserve Bank of India has introduced digital lending regulations that require explicit consent of the borrower, disclosure of the reasons for loan denials, and standardized access to credit reports. These measures bring transparency and consumer rights closer to the “right to explanation” envisioned in fairness research [3]. They mark an important move from opaque model decisions toward procedural fairness.

However, the reforms regulate data handling, not the algorithms themselves. There is no mandate to measure or mitigate disparate impact bias that may persist within model logic. As a result, the guidelines risk equating transparency with fairness, offering procedural visibility without ensuring equitable outcomes. Moreover, awareness gaps and limited enforcement reduce their accessibility for marginalized borrowers, making fairness contingent on literacy rather than law.

5.2 Global Guidelines: Fairness as Principle, Not Practice

At the international level, the World Bank and IMF promote general principles that encourage transparency, non-discrimination, and explainability in credit scoring. These guidelines reflect the same ideas discussed in fairness research but express them in policy language. However, they remain mostly aspirational because there are no strict rules or audits to ensure that institutions actually follow them. Phrases like “models should not discriminate” sound good in theory but are rarely backed by clear definitions or measurable criteria.

As a result, many institutions may focus on appearances rather than real change, offering simple explanations to regulators without addressing deeper issues of bias. This creates what some researchers call “explanation theater,” where fairness becomes more about presentation than accountability.

5.3 Policy Synthesis

Both frameworks share a procedural vision of fairness centered on disclosure, consent, and transparency. This represents progress but remains limited compared to the substantive fairness goals of algorithmic research i.e. equal opportunity, group parity, or counterfactual consistency [1,2]. Policies ensure that individuals can see and challenge decisions but do not guarantee that the underlying distribution of credit access is just.

True progress will require bridging technical and regulatory approaches: embedding fairness metrics and audit requirements into policy, while designing models that align with evolving legal definitions of accountability. Fair credit scoring cannot rely solely on improved models or improved oversight. It depends on their integration. Only when fairness becomes both a property of the algorithm and a mandate of governance can systems claim to be truly equitable.

one concise summary

Paper	Main Focus	Fairness Notion	Key Contribution	Key Limitation
Kusner et al. [1]	Causal modeling of fairness	Counterfactual fairness (individual level)	Introduces a structural causal framework that formalizes fairness via counterfactual predictions for protected attributes.	Requires strong assumptions about the causal data generating process that are computationally expensive in real credit environments.
Kozodoi et al. [2]	Empirical fairness interventions in credit scoring	Group metrics (Eg. : equalized odds)	Benchmarks pre, in, and post-processing interventions on 7 real datasets, showing fairness can improve with limited profit loss.	Focuses on short-term model performance, with limited attention to long-term borrower outcomes or structural inequities.
Chen et al. [3]	Interpretability under fairness constraints	Stability of explanations	Shows how fairness constraints can destabilize feature attributions (SHAP/LIME), highlighting a tension between fairness and explainability.	Primarily diagnostic: documents instability but offers limited methodological guidance on how to design stable, fair explanations.
Hlongwane et al. [4]	Alternative data for thin-file borrowers	Inclusion-oriented fairness	Argues for using non-traditional data (e.g., mobile, regional signals) to extend credit access to excluded populations.	Alternative data may encode sensitive traits, raising privacy and proxy-discrimination concerns that are not fully addressed.
Kozodoi et al. [5]	Sampling bias in credit scoring	Bias reduction via BASL/BE	Proposes Bias-Aware Self-Learning and Bayesian Evaluation to account for unobserved outcomes of rejected applicants.	Methods mitigate statistical bias but do not engage with broader institutional drivers of exclusion or discrimination.
Hurlin et al. [6]	Fairness of deployed bank models	Demographic parity, equal opportunity	Empirically evaluates operational European credit models, linking fairness testing to regulatory practice.	Relies on standard group metrics and largely stops at compliance-oriented recommendations rather than systemic reform.

Table 1: Summary of reviewed works on fairness in credit scoring, highlighting their main focus, fairness notions, and conceptual limitations.

6 Future Directions and Final Remarks

While policy frameworks have started to include fairness principles, much work remains to connect what is measured in algorithms to what is enforced in law. Future research should aim to build unified evaluation systems that can link technical fairness metrics such as demographic parity or equal opportunity with real-world outcomes like credit accessibility and borrower welfare. This would help regulators assess whether fairer models actually lead to fairer results in practice.

Another open question is how fairness changes over time. Most studies evaluate fairness only once, at the time a model is built. Future work should explore how fairness evolves as data, economic conditions, and borrower behavior shift, and how continuous monitoring or “fairness auditing” can prevent bias from resurfacing.

Interdisciplinary collaboration will also be key. Data scientists, economists, and legal experts need to work together to redefine fairness as not just a mathematical property but a shared public goal.

This collaboration could help answer difficult questions such as who decides what fairness means in credit scoring and how can fairness be made measurable without oversimplifying social realities?

Finally, future research should explore participatory and inclusive model design, where affected groups have a voice in how lending models are built and evaluated. Fairness in credit scoring must evolve from being a purely technical adjustment to becoming a long-term institutional commitment, one that ensures lending systems promote social equity as carefully as they manage financial efficiency.

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